

REVIEW - NARRATIVE

Estrogen therapy and breast cancer in randomized clinical trials: a narrative review

Kathy Pan, MD,¹ Sayeh Lavasani, MD,² Aaron K. Aragaki, MS,³ and Rowan T. Chlebowski, MD, PhD¹

Abstract

Importance and Objective: In the Women's Health Initiative (WHI) randomized trial with 10,739 postmenopausal women with prior hysterectomy, conjugated equine estrogen (CEE) alone significantly reduced breast cancer incidence and breast cancer mortality. In contrast, epidemiological studies in a meta-analysis from the Collaborative Group on Hormonal Factors in Breast Cancer (Collaborative Group) with 108,647 breast cancers and the Million Women's Study cohort significantly associated estrogen-alone therapy with higher breast cancer incidence and breast cancer mortality. The Collaborative Group included a meta-analysis of five smaller randomized trials and the WHI randomized trial; however, findings were restricted to the Collaborative Group appendix. Our objective is to facilitate understanding of these discordant results.

Methods: Data sources supporting our review findings include the randomized WHI CEE-alone trial and the meta-analysis of five smaller randomized trials evaluating estrogen alone. We summarize the smaller randomized trials' details of breast cancer relevance and place the findings in clinical context. We review findings of the WHI randomized trial evaluating CEE alone in the context of issues raised by Collaborative Group and the Million Women Study authors. We trace the evolution of the time-from-menopause, "window of opportunity" concept and augment the Collaborative Group meta-analysis by including the most recent WHI findings.

Discussion and Conclusions: Consideration of the smaller randomized trials evaluating estrogen alone with breast cancer signals that the WHI findings of lower breast cancer incidence and lower breast cancer mortality with CEE-alone use are not a "stand-alone" outcome or due to the play of chance. The serial reports of consistent favorable breast cancer findings through 20 years of cumulative follow-up suggest CEE-alone use initiates changes that persist. After full consideration of risks and benefits, randomized trial evidence provides reassurance for postmenopausal women with prior hysterectomy who are close to menopause considering estrogen alone for climacteric symptom management.

Key Words: Breast cancer – Estrogen therapy – Randomized trials – Women's Health Initiative.

The Women's Health Initiative (WHI) randomized placebo-controlled clinical trial evaluating 0.625 mg/d of conjugated equine estrogen (CEE) alone entered 10,739 postmenopausal women with prior hysterectomy, aged 50 to 79 years, from 40 US clinical centers between 1993 and 1998 with continu-

ing ongoing follow-up.¹ Women with prior breast cancer were excluded, and a mammogram not suggestive of cancer was required for entry. Annual mammograms were protocol mandated, and documented compliance was required for annual study medication distribution. Breast cancers were verified by central medication review. Mortality information was enhanced by serial National Death Index queries, which capture 98% of deaths.² After 7.2 years (median) of CEE-alone versus placebo intervention and 20 years (median) of cumulative follow-up, CEE alone, compared with placebo, was associated with a statistically significantly 22% lower breast cancer incidence and was associated with a statistically significantly 40% lower breast cancer mortality.³

In contrast, findings on estrogen therapy influence on breast cancer from the Collaborative Group on Hormonal Factors in Breast Cancer (Collaborative Group)⁴ and the Million Women's Study⁵ substantially differ from the WHI randomized clinical trial results. The Collaborative Group meta-analysis of 58 observational studies included 108,647 postmenopausal women with breast cancer. The principal analyses were described as using "individual participant data from all eligible prospective studies

Received December 17, 2021; revised and accepted March 17, 2022.

From the ¹The Lundquist Institute, Torrance; and ²Division of Medical Oncology & Experimental Therapeutics, City of Hope Comprehensive Cancer Center, Duarte, CA; and the ³Division of Public Health Sciences, Fred Hutchinson Cancer Research Center, Seattle, WA.

Funding/support: The WHI program is funded by the National Heart, Lung, and Blood Institute, National Institutes of Health, US Department of Health and Human Services through contracts HHSN268201600018C, HHSN268201600001C, HHSN268201600002C, HHSN268201600003C, and HHSN268201600004C. K.P. has received support from R25CA203650. Financial disclosure/conflicts of interest: R.T.C. is a consultant to Novartis, AstraZeneca, and Puma and received honorarium from Novartis and AstraZeneca. None of the other authors report any conflict of interest related to this study.

Address correspondence to: Rowan T. Chlebowski, MD, PhD, The Lundquist Institute, 1124 W Carson St, Torrance, CA 90502. E-mail: rowanchlebowski@gmail.com

that had information on type and timing of menopausal hormone therapy” in a nested case-control design.⁴ No information on mammogram screening for eligibility or ongoing serial mammography was provided with hormone therapy duration largely based on retrospective self-report. In this setting, a consistent, statistically significant increase in breast cancer risk with estrogen-alone therapy during years 5 to 14 of use is seen for both equine estrogen (n cases = 1,910; relative risk [RR], 1.32; 95% confidence interval [CI], 1.25-1.39) and estradiol (n cases = 1,563; RR, 1.38; 95% CI, 1.30-1.46). The excess breast cancer risk increased with estrogen therapy use duration and was less than that seen for combined estrogen plus progestin use.⁴

The Million Women Study was a population-based, prospective study with 907,162 postmenopausal women who were free from breast cancer when recruited from 1996 to 2001 from 66 National Health Service screening centers and followed until January 1, 2018.⁵ More than 90% of participants accepted their screening invitation approximately 3 years following recruitment regardless of their hormone therapy use status. During approximately 20 years of follow-up, 7,086 women died of breast cancer (0.8%). Current users of estrogen alone had significant excess breast cancer mortality ($P < 0.0001$); specifically, for current estrogen therapy users for 5 years or more, breast cancer mortality was 35% higher (RR, 1.35; 95% CI, 1.24-1.47).⁵

RANDOMIZED TRIAL RESULTS IN THE COLLABORATIVE GROUP REPORT

Findings from the WHI CEE-alone and WHI CEE plus medroxyprogesterone acetate (MPA) randomized trials were contributed to the Collaborative Group. Findings from the two WHI randomized hormone therapy trials and seven smaller randomized trials with information on hormone therapy and breast cancer were presented as separate trial results and as meta-analyses. The results were restricted to the Collaborative Group appendix (Tables S17 and S18)⁴ as they “did not fit the Collaborative group criteria,” adopted in 2001 when only observational studies were included, of having at least 1,000 breast cancer cases. However, the randomized trial findings were commented on in the Collaborative Group discussion.

The Cooperative Group investigators found five trials that supplied information on breast cancer incidence where findings for estrogen therapy were considered separately from estrogen plus progestin. By design, the trials were smaller and of shorter duration. We focus on the design and details of the smaller randomized trials supplying findings on estrogen therapy use given the discordant findings in the WHI estrogen therapy randomized trial and the Collaborative Group and Million Women Study reports.

Details of smaller estrogen therapy randomized trials with breast cancer findings

We present details of relevance to breast cancer of the five smaller randomized trials with information on breast cancer below. These include study design and study endpoints, participant characteristics, randomization methods, mammography as entry criterion, information on breast cancer risk (family history, reproductive history, BMI, age), breast cancer findings, and summary

of primary study outcomes. Two of the five randomized trials, the Estrogen Replacement and Atherosclerosis (ERA) trial⁶ and the Postmenopausal Estrogen/Progestin Intervention (PEPI) trial,⁷ report only a total of three breast cancer cases. However, their designs provide information on concepts regarding estrogen-alone influence on cardiovascular disease (CVD) and other clinical outcomes in 1990-1996.

The PEPI trial

The PEPI trial (Table 1) was started in 1989 and included healthy postmenopausal women, aged 45 to 64 years, with or without a uterus. Participants were randomized to placebo, CEE-alone, and CEE plus three different progestin regimens. Endpoints were circulatory markers or mediators of CVD. The inclusion of women with a uterus indicates that the issue of estrogen influence on endometrial cancer was considered an open question. The positive influence on study endpoints, reported in 1995,⁷ was suggestive of CVD benefit.

ERA trial

The ERA trial (Table 2), started in 1995, included women 55 years or older with angiographically verified coronary disease with endpoint of change in coronary artery diameter.⁶ Randomization was to placebo, CEE alone, and CEE plus MPA. As 44% of women with a uterus were randomized to CEE alone, the endometrial cancer issue was considered unsettled. With a mean age of 65.8 years, “window of opportunity” was not yet a consideration when findings were reported in 2000. The negative findings on coronary artery change from both interventions led to a conclusion against use of “estrogen replacement” for CVD benefit.⁶

Women's Estrogen for Stroke Trial

The Women's Estrogen for Stroke Trial (WEST) (Table 3), initiated in 1993, entered postmenopausal women with a recent stroke or transient ischemic attack within 90 days of entry. Death from any cause and nonfatal stroke were primary

TABLE 1. The PEPI trial

Participants: 875 Healthy postmenopausal women with or without a uterus. Aged 45-64 y, with no hormone therapy contraindication
Setting: Recruitment from seven US clinical centers between 1989 and 1991
Intervention: Randomization to (1) placebo; (2) CEE alone 0.625 mg/d; (3-5) CEE 0.625 mg/d plus three different progestin regimens
Randomization method: A blocked randomization was determined by computer program, not stated whether centrally or at clinical sites
Breast cancer risk: Breast cancer family history (not available), no prior breast cancer, mammogram at entry and annually required, BMI (not available), mean age 56.1 y
Primary study outcome: Endpoints were high-density lipoprotein cholesterol (HDL-C), systolic blood pressure, serum insulin, and fibrinogen
Study conduct: Assessment after 3 y of intervention
Breast cancer results: Breast cancers in 1 of 175 in the placebo group and 1 of 174 in the CEE-alone group.
Primary study conclusion: Estrogen alone or with a progestin improves lipoproteins and lowers fibrinogen levels. Estrogen alone increases HDL-C, but endometrial hyperplasia restricts use to women without a uterus

BMI, body mass index; CEE, conjugated equine estrogen; PEPI, Postmenopausal Estrogen/Progestin Intervention.

“Hormone therapy was not associated with excess breast cancer risk, but short follow-up and limited numbers cannot exclude excess with longer use.”

TABLE 2. The ERA trial

Participants: Postmenopausal women with angiographically verified coronary disease were eligible. Prior breast or endometrial cancer or prior cardiovascular disease history or risk factors and current hormone therapy use were exclusions. Of the 815 women screened, 309 (38%) enrolled

Setting: Recruitment from six US clinical sites in 1995 and 1996

Intervention: Randomization to (1) placebo; (2) CEE alone 0.625 mg/d; (3) CEE 0.625 mg/d plus 2.5 mg/d MPA

Randomization: A permuted-block randomization procedure was used after stratification by lipid-lowering therapy use and clinical site

Breast cancer risk: Breast cancer family history (not available), no prior breast cancer, mammograms (no information), BMI (not available), (mean) age 56.1 y

Primary study outcome: Baseline vs follow-up mean minimal coronary-artery diameter with other angiographic outcomes within each subject

Study conduct: Angiographic data at baseline and angiographic data after 3.2 ± 0.6 y (mean) cumulative follow-up, available for 248 of the 309 subjects

Results: Breast cancers in 0 of 105 placebo group vs 1 of 100 in CEE group ($P = 0.35$)

Primary study conclusion: Neither estrogen alone nor estrogen plus progestogen affected the progression of coronary atherosclerosis in women with established disease

BMI, body mass index; CEE, conjugated equine estrogen; ERA, Estrogen Replacement and Atherosclerosis; MPA, medroxyprogesterone acetate. Of 100 women in the CEE-alone group, 56 had prior hysterectomy. In women with a uterus in that group, heavy or persistent bleeding was seen in 56%.

outcomes (Figure 3 in Viscoli et al⁸). Mean age of participants was 71 years. Randomization was to placebo or estradiol 17 β . Breast cancer was acknowledged as a potential adverse effect. Prior breast cancer was an exclusion, clinical breast examination and mammography were required before entry, and mammography was performed annually. There were incident breast cancers in 5 of 327 in the placebo group and 5 of 337 in the estradiol-17 β group (RR, 1.0; 95% CI, 0.3-3.5). The primary study conclusion, reported in 2001, was that estradiol-17 β did not reduce mortality or stroke recurrence.⁸

TABLE 3. The WEST trial

Participants: 664 Postmenopausal women 44 y of age with an ischemic stroke or transient ischemic attack within 90 d were randomized. Prior breast or endometrial cancer and venous thromboembolic event on hormone therapy were exclusions

Setting: Recruited from 21 US hospitals between 1993 and 1998

Intervention: Randomization to 1 mg/d estradiol-17 β or placebo regardless of hysterectomy status; 44% of participants had a prior hysterectomy

Randomization: A primary list of computer-generated random treatment assignments was stored at the investigational pharmacy at Yale-New Haven Hospital. Randomization was stratified according to clinical center (21 hospitals) and baseline risk group

Breast cancer risk: Breast cancer family history (not available), no prior breast cancer, mammogram at entry and annually required, BMI mean 28.7 kg/m², mean age 72 y

Primary study outcome: Death from any cause or nonfatal stroke; secondary outcomes were transient ischemic attack and nonfatal myocardial infarction

Study conduct: 2.8 y (mean) intervention and same cumulative follow-up

Breast cancer results: Breast cancers in 5 of 327 in placebo group vs 5 of 337 in estradiol-17 β group (RR, 1.0; 95% CI, 0.3-3.5)

Primary study conclusion: "Estradiol-17 β alone does not reduce mortality or the recurrence of stroke in postmenopausal women with cerebrovascular disease"

BMI, body mass index; WEST, Women's Estrogen for Stroke Trial. Both the National Institutes of Health and the Federal Drug Administration do not allow use of the terms *Hormone Replacement Therapy* or *HRT*. Those terms suggest menopause is a hormone deficiency state where an established intervention can reverse the condition. That is not the case with respect to menopausal hormone therapy.

TABLE 4. DOPS trial participants

Participants: 1,060 Healthy women aged 45 to 58 y, recently postmenopausal or perimenopausal women and women with prior hysterectomy, both with defined follicle-stimulating hormone levels, were eligible. Prior bone disease and prior cancer were exclusions

Setting: DOPS trial participants in 1990-1993

Intervention: Hormone therapy vs no therapy (open-label): (1) women with prior hysterectomy: 2 mg/d estradiol-17 β (n = 192), (2) women with a uterus: estradiol-17 β plus norethisterone acetate (n = 814), regimens continued through approximately 10 y

Randomization method: Participants were stratified according to center and randomly allocated to hormone therapy or no therapy in blocks of 10, using sealed envelopes

Breast cancer risk: Breast cancer family history (not available), no prior breast cancer, mammograms (no information), BMI mean 25.3 (SD, 4.3) kg/m², mean age 49.5 y

Primary study outcome: The primary endpoint was a composite of death, admission to hospital for heart failure, and myocardial infarction. Breast cancer incidence was monitored as a safety endpoint

Study conduct: Intervention stopped after approximately 11 y with cumulative follow-up through 16 y

Breast cancer results: Breast cancers in 9 of 97 in placebo group vs breast cancer in 6 of 95 in estradiol-17 β group (HR, 0.63; 95% CI, 0.23-1.78). In addition, a statistically significant reduction in the predefined composite endpoint of mortality or breast cancer in the estradiol-17 β group (HR, 0.47; 95% CI, 0.18-0.97)

Primary study conclusion: Women receiving hormone therapy early after menopause had a significantly reduced risk of mortality, heart failure, or myocardial infarction, without any apparent increase in risk of cancer, venous thromboembolism, or stroke

BMI, body mass index; DOPS, Danish Osteoporosis Prevention Study. Both the National Institutes of Health and the Federal Drug Administration do not allow use of the terms *hormone replacement therapy* or *HRT*. Those terms suggest menopause is a hormone deficiency state where an established intervention can reverse the condition. That is not the case with respect to menopausal hormone therapy.

Danish Osteoporosis Prevention Study

A trial conducted in the Danish Osteoporosis Prevention Study (DOPS) population (Table 4), started in 1990, randomized healthy women who were recently (within approximately 7 months) postmenopausal or had perimenopausal symptoms with supportive serum follicle-stimulating hormone values to no therapy (open label) or estradiol-alone if prior hysterectomy or estradiol plus norethisterone if the woman had a uterus.⁹ Prior breast cancer was an exclusion; no information on mammography was provided. Hormone therapy intervention was approximately 10 years with follow-up through 16 years. The primary endpoint was a composite of death, heart failure requiring hospitalization, and myocardial infarction. There were 9 of 97 breast cancers in the placebo group versus 6 of 95 breast cancers in the estradiol group (hazard ratio [HR], 0.63; 95% CI, 0.23-1.78). In addition, there was a statistically significant reduction in the predefined composite endpoint of mortality or breast cancer in the estradiol group (HR, 0.47; 95% CI, 0.18-0.97). The positive effect on the primary study composite endpoint was presented comparing the combined findings in the estradiol-alone group plus those in the estradiol plus norethisterone group to findings in the no-treatment (rather than placebo) control group (Figure 2 in Schierbeck),⁹ raising some issues.^{10,11}

Estrogen for the Prevention of Re-Infarction Trial

The Estrogen for the Prevention of Re-Infarction Trial (ESPRIT) (Table 5), initiated in 1996, randomized 1,017 women with or

TABLE 5. The ESPRIT trial

Participants: Women aged 50-69 y with an intact uterus surviving their first myocardial infarction were eligible. Prior breast, endometrial, or ovarian cancer or prior hormone therapy use was excluded; 140 of 513 with hysterectomy were in active therapy group
Setting: 1,017 Women from 35 hospitals in England and Wales in 1996-2000
Intervention: Estradiol valerate 2 mg or placebo for 2 y
Randomization: Centralized randomization was stratified by hospital. Schering AG prepared numbered packages of corresponding treatments blinded to the receiving hospital
Breast cancer risk: Breast cancer family history (not available), no prior breast cancer, mammography (no information), BMI 26.8 (SD, 5.2) kg/m ² ; mean age 63 y
Primary study outcomes: For the current postintervention report, outcomes were death (all-cause, cardiac disease, stroke, or cancer) and cancer incidence (any, breast cancer or endometrial cancer). Breast cancer verification was by International Disease Codes after linkage to the UK cancer record
Study conduct: 2-y intervention and 14-y cumulative follow-up
Breast cancer results: Breast cancers in 15 of 504 in the placebo group vs breast cancers in 7 of 513 in the estradiol valerate group (HR, 0.47; 95% CI, 0.19-1.15)
Primary study conclusion: "Results suggest that unopposed estrogen may be used safely by women with an intact uterus surviving a first myocardial infarction"

BMI, body mass index; ESPRIT, Estrogen for the Prevention of Re-Infarction Trial. During intervention, vaginal bleeding occurred in 56% of women in the active therapy group who had a uterus (208 of 373). This is a conclusion that current findings on estrogen therapy and endometrial cancer do not support.

without a uterus, surviving their first myocardial infarction, aged 50 to 69 years (mean, 63 years), from 35 hospitals in England and Wales to estradiol valerate or placebo for 2 years.¹² Prior breast cancer was an exclusion. No information on mammography was provided. Primary outcomes for the 14-year cumulative follow-up were death and cancer incidence (any, breast cancer, endometrial). There were 15 of 504 breast cancers in the placebo group versus 7 of 513 breast cancers in the estradiol valerate group (HR, 0.47; 95% CI, 0.19-1.15). The primary study conclusion was that "unopposed estrogen may be used safely by women with an intact uterus surviving a first myocardial infarction." This is a conclusion that current findings on estrogen therapy and endometrial cancer do not support.

Summary

As the PEPI, ERA, WEST, DOPS, and ESPRIT randomized trials all began accrual between 1989 and 1996 trials, their study designs reflect concepts regarding menopausal hormone therapy

when the WHI randomized trials were started in 1993. Four of the five trials were willing to randomize women with a uterus to estrogen therapy because the need for progestin to mitigate endometrial cancer risk associated with estrogen therapy was not quite settled. Only one of the trials addressed a hypothesis in recently postmenopausal women.⁹ In three trials, CVD,⁶ myocardial infarction,¹² and stroke⁸ were eligibility requirements with mean age at entry of 63, 66, and 71 years. These designs reflected the then-current findings from consensus of observational study results that estrogen therapy was associated with lower CVD events^{13,14} and fractures¹³ and should be evaluated for secondary prevention. Finally, there was emerging concern about estrogen therapy and higher breast cancer risk.¹⁵

Breast cancer incidence and mortality in estrogen therapy randomized trials

Combining results from PEPI, ERA, WEST, DOPS, and ESPRIT trials,^{6-9,12} there were 17 breast cancers in 1,220 women in the estrogen therapy groups and 28 breast cancers in 1,206 women in the placebo/no therapy groups (RR, 0.61; 95% CI, 0.34-1.09; 2P = 0.15).⁴ Three additional randomized trials evaluating estrogen therapy with breast cancer findings have been identified.¹⁶ Across these three trials,¹⁷⁻¹⁹ the breast cancer distribution by therapy group assignment would not materially influence the results of the smaller randomized trials. In summary, breast cancer findings from the five smaller randomized trials included in the Collaborate Group meta-analysis are consistent with the results of the WHI randomized, placebo-controlled trial with 10,739 participants, 20 years of follow-up, and 534 incident breast cancers.

The WHI CEE-alone findings were reported in the Cooperative Group⁴ when there were 384 incident breast cancers after 13 years of follow-up through September 2010 (RR, 0.79; 95% CI, 0.65-0.97; 2P = 0.02)²⁰ and when combined with findings from the smaller trials resulted in a combined RR of 0.77 (95% CI, 0.64-0.93; 2P = 0.01) (Table 6).

Further information on the evolution of the breast cancer findings in the WHI CEE-alone trial follows. In the initial report of findings in the WHI trial evaluating CEE-alone in 2004, at the end of the 7.2-year intervention period, there were fewer breast

TABLE 6. Summary results from randomized trials of estrogen therapy vs placebo and breast cancer incidence and breast cancer mortality

Results from collaborative group analyses, breast cancer incidence ^a	
	RR (95% CI)
Subtotal/five smaller estrogen therapy trials ^b	0.61 (0.34-1.09), 2P = 0.15
WHI CEE-alone trial ^c	0.79 (0.65-0.97), 2P = 0.02
WHI CEE-alone trial plus five smaller estrogen therapy trials ^d	0.77 (0.64-0.93), 2P = 0.01
Results from WHI CEE alone trial update, breast cancer incidence, and mortality ^e	
	HR (95% CI)
WHI CEE-alone trial and breast cancer incidence ^e	0.78 (0.65-0.93), P = 0.005
WHI CEE-alone trial and breast cancer mortality ^f	0.60 (0.37-0.97), P = 0.04

CEE, conjugated equine estrogen; WHI, Women's Health Initiative.

^aAdded to and adopted from supplement to *Lancet* 2019; online August 29, 2019, [http://dx.doi.org/10.1016/S0140-6736\(19\)31709-X](http://dx.doi.org/10.1016/S0140-6736(19)31709-X).

^bSubtotal/five smaller trials with 2,426 participants and 45 breast cancers.

^cWHI CEE-alone trial with 10,739 participants and 384 breast cancers.

^dMeta-analysis of WHI CEE-alone trial plus five smaller trials with 13,165 participants and 429 breast cancers.

^eUpdate of WHI CEE-alone trial for breast cancer incidence with 10,739 participants and 534 breast cancers.³

^fUpdate of WHI CEE-alone trial for breast cancer mortality with 10,739 participants, 76 with breast cancer mortality.³

cancers in the intervention group, but the finding was not statistically significant.^{21,22} With longer cumulative 11.8-year (median) follow-up, CEE-alone use was seen to significantly reduce breast cancer incidence (HR, 0.77; 95% CI, 0.62-0.95; $P = 0.02$) and reduce breast cancer mortality (HR, 0.37; 95% CI, 0.13-0.91; $P = 0.03$).²³ The findings were maintained through additional postintervention follow-up^{20,24} and were similar in the 1,616 Black women participating in the trial.²⁵ Most recently, with cumulative 20 years (median) of follow-up including 13 years of postintervention follow-up, CEE-alone use continued to decrease breast cancer incidence (HR, 0.78; 95% CI, 0.65-0.93; $P = 0.005$) and decrease breast cancer mortality (HR, 0.60; 95% CI, 0.37-0.97; $P = 0.04$) (Table 6).³ Findings were similar regardless of age decade, bilateral oophorectomy status, and prior hormone therapy use. In addition, the significant reduction in poor-prognosis, estrogen receptor–positive, progesterone receptor–negative breast cancers (HR, 0.44; 95% CI, 0.27-0.74) supports the mortality results.³

Breast cancer and estrogen alone: randomized trials versus observational studies

Although there are considerable differences comparing randomized clinical trials to observational studies and meta-analyses of observational studies, the contrasting findings regarding estrogen-alone use and breast cancer incidence and mortality cannot be completely explained. Women in the WHI randomized trial evaluating CEE-alone were older with longer time from menopause to hormone use (gap time). Although age and gap time alter estrogen therapy influence on other outcomes including coronary heart disease²⁶ and all-cause mortality,²⁴ for breast cancer in the WHI CEE-alone trial, there was no statistically significant interaction by age decade (50-59, 60-69, 70-79 years), prior hormone therapy use (yes/no), or gap time (<5, ≥5 years); in all subgroups, HRs for breast cancer incidence were less than 1.³ The lack of information on mammography is a potential confounder of the cooperative group findings⁴ but not those of the Million Women Study because mammography information was available and compliance was high.⁵ Finally, reliance on self-reports for hormone therapy duration represents an analytical issue for many observational studies.

Addressing the contrasting results regarding breast cancer incidence in observational studies compared with the randomized trials, the authors of the Collaborative Group meta-analysis suggested the decrease in breast cancer with use of estrogen therapy in randomized trials arose “mainly by the play of chance” augmented by increased breast density with “reduction in mammographic sensitivity.”⁴ However, although CEE alone increases breast density to a degree, when mammography performance was formally evaluated, CEE alone had no adverse influence on breast cancer detection, a finding published 9 years earlier than the Collaborative Group report.^{4,27} In addition, the statistically significant 40% reduction in deaths from breast cancer associated with CEE alone in the WHI randomized trial³ provides definitive evidence against a screening artifact; detection delay would likely increase, rather than decrease, breast cancer mortality. The play of chance to explain the CEE-alone finding on breast can-

cer incidence (HR, 0.78; 95% CI, 0.65-0.93; $P = 0.005$)³ is also statistically inconsistent (probability calculated as $P < 0.0001$) with the Collaborative Group observational analyses for current estrogen therapy use (RR, 1.33; 95% CI, 1.28-1.38).⁴ In addition, there are plausible biological hypotheses regarding estrogen and mammary tumor growth, including estrogen influence on estrogen-dependent apoptosis^{28,29} and effects of estrogen withdrawal on tumor growth.³⁰ Finally, clinical evidence supports use of estradiol (6 mg/d) as therapy for postmenopausal advanced aromatase inhibitor–resistant, hormone receptor–positive breast cancer.³¹ However, the 6 mg/d estradiol dose used would be considered high-dose compared with low-dose estradiol at 0.5 to 1 mg/d, currently recommended for vasomotor symptom management.³²

Evolution of the timing hypothesis

Following the initial reports of the WHI randomized trials in 2002³³ and 2004,²¹ the concept emerged that timing from menopause may represent a “window of opportunity” to see estrogen benefits.³⁴ In response, the Kronos Early Estrogen Prevention Study (KEEPS) examined the influence of timely (within 3 years of last menstrual period [2 years from menopause]) estrogen plus progestin therapy on cardiovascular health³⁵; however, although expected favorable influence on menopausal symptoms was seen, there was no intervention effect on the primary endpoint of carotid artery intima-media thickness.³⁶

Subsequently, findings for estrogen therapy use initiated 5 or more years from menopause “gap time” have been examined in the Million Women Study where no increase in breast cancer was seen when obesity was included in the analysis (RR, 0.91; 95% CI, 0.73-1.14).³⁷ Similarly, early in the WHI CEE-alone trial, breast cancer incidence was lower in women with a gap time of 5 or more years.²³ With longer cumulative follow-up in the CEE-alone trial, gap time was no longer a significant interaction, indicating that additional mechanisms, unrelated to gap time, may mediate the lower breast cancer incidence and mortality associated with use of CEE alone.³

Examination of more global mortality outcomes in the WHI hormone therapy trials by 10-year groups found favorable outcomes limited to women aged 50 to 59 years with prior hysterectomy for CEE-alone use. In that group, through cumulative follow-up, in addition to the breast cancer benefit,³ coronary heart disease was 35% lower; all cancers, excluding nonmelanoma skin cancers, were 20% lower²⁶; and all-cause mortality was 21% lower, with all findings statistically significant.^{24,38} Thus, the “window of opportunity” has become a decade of opportunity. These benefits were not seen with CEE plus MPA use.

Study limitations include the disparate study designs of the five smaller randomized trials with differing estrogens evaluated (CEE, estradiol-17β, estradiol valerate) and differing primary outcomes. Against this background, the consistent breast cancer findings seen in those trials to findings in the large WHI trial are perhaps more remarkable. There are limitations inherent in narrative reviews, shared by this report, as the format provides some flexibility to the authors regarding their selection and interpretation of the reviewed evidence. However, the authors

are well-credentialed in this area, leading a WHI Clinical Center (R.T.C.), conducting WHI analyses (A.K.A.), and publishing WHI findings (all authors).

CONCLUSIONS

Consideration of the smaller randomized trials evaluating estrogen therapy with breast cancer outcomes provides insights to the concepts regarding menopausal hormone therapy when the WHI randomized hormone therapy trials were initiated in 1993. In addition, although the number of events was limited, the smaller studies provide a signal that the WHI findings in postmenopausal women with prior hysterectomy of lower breast cancer incidence and lower breast cancer mortality with CEE-alone use are not a “stand-alone” outcome or due to the play of chance. The serial reports of consistent findings of a favorable influence on breast cancer developing through 7.2 years of intervention and 13 years of postintervention follow-up suggest CEE alone initiates changes that influence breast cancer that persists. After full consideration of risks and benefits, the available randomized trial evidence provides reassurance for postmenopausal women with prior hysterectomy who are close to menopause considering estrogen therapy for climacteric symptom management.

Acknowledgments: The authors thank the WHI investigators, staff, and the trial participants for their outstanding dedication and commitment.

Women's Health Initiative Investigators: Program Office: (National Heart, Lung, and Blood Institute, Bethesda, MD) Jacques Roscoe, Shari Ludlum, Dale Burden, Joan McGowan, Leslie Ford, and Nancy Geller. Clinical Coordinating Center: (Fred Hutchinson Cancer Research Center, Seattle, WA) Garnet Anderson, Ross Prentice, Andrea LaCroix, and Charles Kopperberg. Investigators and Academic Centers: (Brigham and Women's Hospital, Harvard Medical School, Boston, MA) JoAnn E. Manson; (MedStar Health Research Institute/Howard University, Washington, DC) Barbara V. Howard; (Stanford Prevention Research Center, Stanford, CA) Marcia L. Stefanick; (The Ohio State University, Columbus, OH) Rebecca Jackson; (University of Arizona, Tucson/Phoenix, AZ) Cynthia A. Thompson; (University at Buffalo, Buffalo, NY) Jean Wactawski-Wende; (University of Florida, Gainesville/Jacksonville, FL) Marian Limacher; (University of Iowa, Iowa City/Davenport, IA) Robert Wallace; (University of Pittsburgh, Pittsburgh, PA) Lewis Kuller; (The Lundquist Institute, Torrance, CA) Rowan T. Chlebowski; (Wake Forest University School of Medicine, Winston-Salem, NC) Sally Shumaker.

REFERENCES

- Anderson GL, Manson J, Wallace R, et al. Implementation of the Women's Health Initiative study design. *Ann Epidemiol* 2003;13(9 suppl):S5-S17.
- Rich-Edwards JW, Corsano KA, Stampfer MJ. Test of the National Death Index and Equifax Nationwide Death Search. *Am J Epidemiol* 1994; 140:1016-1019.
- Chlebowski RT, Anderson GL, Aragaki AK, et al. Association of menopausal hormone therapy with breast cancer incidence and mortality during long-term follow-up of the Women's Health Initiative randomized clinical trials. *JAMA* 2020;324:369-380.
- Collaborative Group on Hormonal Factors in Breast Cancer. Type and timing of menopausal hormone therapy and breast cancer risk: individual participant meta-analysis of the worldwide epidemiological evidence. *Lancet* 2019;394:1159-1168.
- Beral V, Peto R, Pirie K, Reeves G. Menopausal hormone therapy and 20-year breast cancer mortality. *Lancet* 2019;394:1139.
- Herrington DM, Reboussin DM, Brosnihan KB, et al. Effects of estrogen replacement on the progression of coronary-artery atherosclerosis. *N Engl J Med* 2000;343:522-529.
- Effects of estrogen or estrogen/progestin regimens on heart disease risk factors in postmenopausal women. The Postmenopausal Estrogen/Progestin Interventions (PEPI) Trial. The Writing Group for the PEPI Trial. *JAMA* 1995; 273:199-208.
- Viscoli CM, Brass LM, Kernan WN, Sarrel PM, Suissa S, Horwitz RI. A clinical trial of estrogen-replacement therapy after ischemic stroke. *N Engl J Med* 2001;345:1243-1249.
- Schierbeck LL, Rejnmark L, Tofteng CL, et al. Effect of hormone replacement therapy on cardiovascular events in recently postmenopausal women: randomised trial. *BMJ* 2012;345:e6409.
- Rossouw JE, Manson JE, Kaunitz AM, Stefanick ML. Study had insufficient power to investigate safety. *BMJ* 2012;345:e8146; author reply e64.
- Marjoribanks J, Farquhar C, Roberts H, Lethaby A. Trial does not change the conclusions of Cochrane review of long term hormone therapy for perimenopausal and postmenopausal women. *BMJ* 2012;345:e8141; author reply e64.
- Cherry N, McNamee R, Heagerty A, Kitchener H, Hannaford P. Long-term safety of unopposed estrogen used by women surviving myocardial infarction: 14-year follow-up of the ESPRIT randomised controlled trial. *BJOG* 2014; 121:700-705 discussion 5.
- Grodstein F, Stampfer MJ, Falkeborn M, Naessen T, Persson I. Postmenopausal hormone therapy and risk of cardiovascular disease and hip fracture in a cohort of Swedish women. *Epidemiology* 1999;10:476-480.
- Grodstein F, Stampfer MJ, Colditz GA, et al. Postmenopausal hormone therapy and mortality. *N Engl J Med* 1997;336:1769-1775.
- Design of the Women's Health Initiative clinical trial and observational study. The Women's Health Initiative Study Group. *Control Clin Trials* 1998;19:61-109.
- Hemminki E, McPherson K. Impact of postmenopausal hormone therapy on cardiovascular events and cancer: pooled data from clinical trials. *BMJ* 1997; 315:149-153.
- Genant HK, Baylink DJ, Gallagher JC, Harris ST, Steiger P, Herber M. Effect of estrone sulfate on postmenopausal bone loss. *Obstet Gynecol* 1990; 76:579-584.
- Gallagher JC, Kable WT, Goldgar D. Effect of progestin therapy on cortical and trabecular bone: comparison with estrogen. *Am J Med* 1991;90:171-178.
- Speroff L, Rowan J, Symons J, Genant H, Wilborn W. The comparative effect on bone density, endometrium, and lipids of continuous hormones as replacement therapy (CHART study). A randomized controlled trial. *JAMA* 1996;276:1397-1403.
- Chlebowski RT, Rohan TE, Manson JE, et al. Breast cancer after use of estrogen plus progestin and estrogen alone: analyses of data from 2 Women's Health Initiative randomized clinical trials. *JAMA Oncol* 2015;1:296-305.
- Anderson GL, Limacher M, Assaf AR, et al. Effects of conjugated equine estrogen in postmenopausal women with hysterectomy: the Women's Health Initiative randomized controlled trial. *JAMA* 2004;291:1701-1712.
- Stefanick ML, Anderson GL, Margolis KL, et al. Effects of conjugated equine estrogens on breast cancer and mammography screening in postmenopausal women with hysterectomy. *JAMA* 2006;295:1647-1657.
- Anderson GL, Chlebowski RT, Aragaki AK, et al. Conjugated equine oestrogen and breast cancer incidence and mortality in postmenopausal women with hysterectomy: extended follow-up of the Women's Health Initiative randomised placebo-controlled trial. *Lancet Oncol* 2012;13:476-486.
- Manson JE, Aragaki AK, Rossouw JE, et al. Menopausal hormone therapy and long-term all-cause and cause-specific mortality: the Women's Health Initiative randomized trials. *JAMA* 2017;318:927-938.
- Chlebowski RT, Barrington W, Aragaki AK, et al. Estrogen alone and health outcomes in black women by African ancestry: a secondary analyses of a randomized controlled trial. *Menopause* 2017;24:133-141.
- Manson JE, Chlebowski RT, Stefanick ML, et al. Menopausal hormone therapy and health outcomes during the intervention and extended poststopping phases of the Women's Health Initiative randomized trials. *JAMA* 2013; 310:1353-1368.
- Chlebowski RT, Anderson G, Manson JE, et al. Estrogen alone in postmenopausal women and breast cancer detection by means of mammography and breast biopsy. *J Clin Oncol* 2010;28:2690-2697.
- Jordan VC, Ford LG. Paradoxical clinical effect of estrogen on breast cancer risk: a “new” biology of estrogen-induced apoptosis. *Cancer Prev Res (Phila)* 2011;4:633-637.
- Santen RJ, Yue W. Cause or prevention of breast cancer with estrogens: analysis from tumor biologic data, growth kinetic model and Women's Health Initiative study. *Climacteric* 2019;22:3-12.
- Howell A, Cuzick J. Oestrogen and breast cancer: results from the WHI trial. *Lancet Oncol* 2012;13:437-438.

31. Ellis MJ, Gao F, Dehdashti F, et al. Lower-dose vs high-dose oral estradiol therapy of hormone receptor–positive, aromatase inhibitor–resistant advanced breast cancer: a phase 2 randomized study. *JAMA* 2009;302:774-780.
32. Pinkerton JV. Hormone therapy for postmenopausal women. *N Engl J Med* 2020;382:446-455. doi:10.1056/NEJMcp1714787
33. Rossouw JE, Anderson GL, Prentice RL, et al. Risks and benefits of estrogen plus progestin in healthy postmenopausal women: principal results from the Women's Health Initiative randomized controlled trial. *JAMA* 2002;288:321-333.
34. Grodstein F, Manson JE, Stampfer MJ. Hormone therapy and coronary heart disease: the role of time since menopause and age at hormone initiation. *J Womens Health (Larchmt)* 2006;15:35-44.
35. Harman SM, Brinton EA, Cedars M, et al. KEEPS: the Kronos Early Estrogen Prevention Study. *Climacteric* 2005;8:3-12.
36. Miller VM, Naftolin F, Asthana S, et al. The Kronos Early Estrogen Prevention Study (KEEPS): what have we learned? *Menopause* 2019;26:1071-1084.
37. Beral V, Reeves G, Bull D, Green J, Million Women Study Collaborators. Breast cancer risk in relation to the interval between menopause and starting hormone therapy. *J Natl Cancer Inst* 2011;103:296-305.
38. Prentice RL, Aragaki AK, Chlebowski RT, et al. Randomized trial evaluation of the benefits and risks of menopausal hormone therapy among women 50-59 years of age. *Am J Epidemiol* 2021;190:365-375.